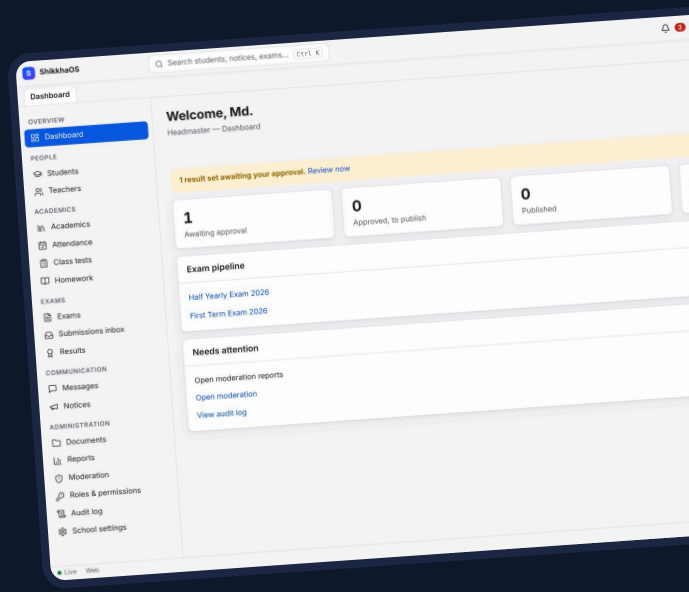
**ShikkhaOS**
SCHOOL OPERATING SYSTEM

# One school. One system.

A single, secure digital platform for administration, teaching, examinations and communication — built for our school.




**Project Proposal**
For the School Managing Committee

**November 2026**
Version 1.0 — for decision

**Platforms**
Windows desktop · Web · Android


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# What is inside

Read pages 3–7 for the idea, pages 8–13 to see the system working, and pages 14–19 for safety, status and the decisions we ask for.


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> **“One School. One System. One Connected Community.”**

EXECUTIVE SUMMARY

# What we are proposing

ShikkhaOS replaces scattered registers, spreadsheets and chat groups with one connected school system that every teacher, the Register, the Headmaster and every student can use — each seeing only what they should.

| 3 | 4 | 7 | 1 |
|---|---|---|---|
| apps: desktop, web, Android | user roles with separate powers | safe steps from marks to published result | permanent record of every change |

### What the school gets

- ✔ Marks entered once, checked by the Register, approved by the Headmaster, then published to students.
- ✔ Daily attendance, class tests and homework in the same place.
- ✔ Notices and messages that reach the right people — with student-safety rules built in.
- ✔ Reports, report cards and certificates generated from the same data.

### What the committee decides

This document shows the real system working, explains how it protects students and results, and asks for six decisions (last pages):

1. Approve a **pilot** in one class.
2. Choose **where the system runs**.
3. Confirm **who may publish results**.
4. Nominate a **system owner** and training time.
5. Confirm the **messaging policy**.
6. Confirm **data handling** and backups.

**About the pictures.** Every screen in this document is a real screenshot of the working system. The names, marks and notices shown are invented demonstration data — no real student information is used.

WHY CHANGE

# Common problems — and what changes

These are problems schools typically face when work lives on paper and in chat groups. We would confirm which apply here during the pilot.

| Typical problem today | With ShikkhaOS |
|---|---|
| Marks are written on sheets, copied to registers and re-typed for the result sheet; small errors slip through. | Teachers enter marks once. The system checks the range (0 to full marks), totals them and calculates grades and positions. |
| Nobody can tell who changed a mark, or when. | Every change is recorded with who, what and when. Submitted marks are locked; corrections go through a visible request. |
| Results are delayed while sheets travel between teachers, Register and Headmaster. | Each person sees exactly what is waiting for them. The Headmaster sees pending approvals on one screen. |
| Students learn results late or by word of mouth. | When the Headmaster publishes, students are notified and see their own result and report card — no one else's. |
| Attendance registers are checked rarely; absences are noticed late. | Teachers take attendance in seconds. Attendance percentages are always up to date for each student and class. |
| Notices and questions go through informal chat groups with no school control. | Official notices and school-managed messaging. Private chat between students is not available; reports and moderation exist. |
| Staff access to records is informal. | Role-based access: a teacher sees only their own classes; the Register and Headmaster have different powers. |

### Built for one school

ShikkhaOS is a private system for our school only — not a shared product. Our data stays under the school's control.

### No parent accounts (first release)

Students use the app. Guardian contact details are stored as records only; parents do not log in during this first release.

THE SYSTEM

# One school server, three ways in

Everyone uses the same data through the app that suits them. Nothing is typed twice.

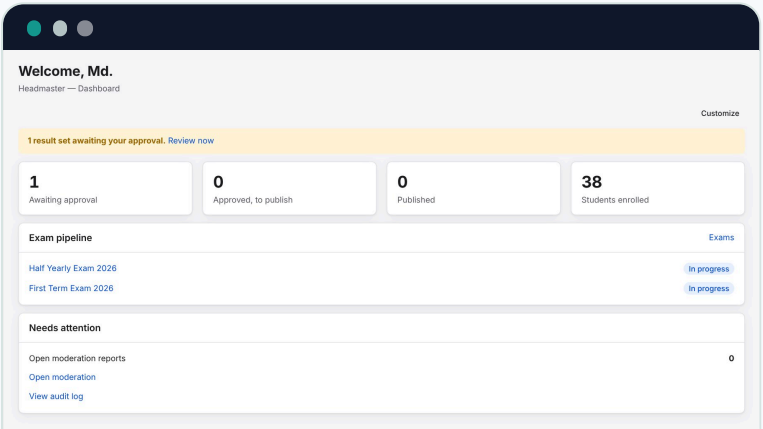
**Windows desktop app**
Headmaster · Register · Teachers

**Web app (any browser)**
All roles

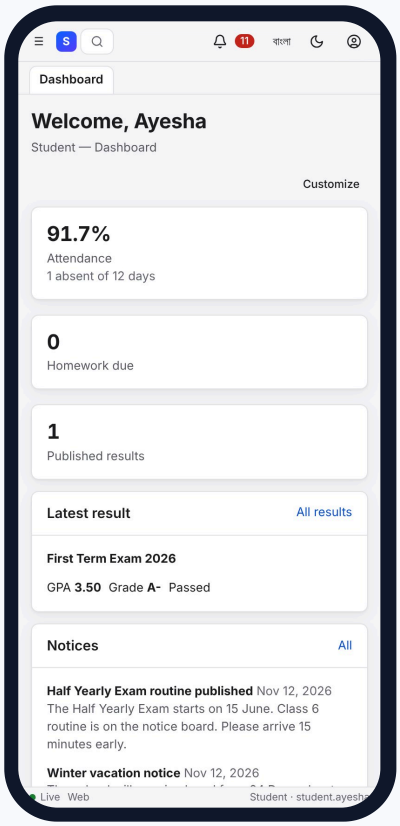
**Android app**
Students

⇅

**ShikkhaOS school server**
Rules · permissions · database · audit record




Headmaster dashboard (desktop/web): what needs attention, at a glance.




Student view on a phone

## Desktop

A proper Windows application for staff: quick search (Ctrl+K), tabs, notifications, light/dark theme.

## Web

Works in any modern browser with the same screens and rules — handy on a shared office computer.

## Android

A student app for results, attendance, homework, notices and messages, with notifications.

PEOPLE AND POWERS

# Who does what

Four kinds of users. What each person can do is set by the school, and enforced by the server — hiding a button is never the only protection.

## Headmaster

- School-wide dashboard and reports
- Authorizes exams; **approves and publishes** results
- Sets who may do what (permissions)
- Official announcements, moderation, audit log

## Register

- Admissions, student and teacher records
- Sets up exams; **checks marks** from teachers
- Requests corrections; compiles results
- Certificates, report cards, documents
- Publishes only if the Headmaster delegates it

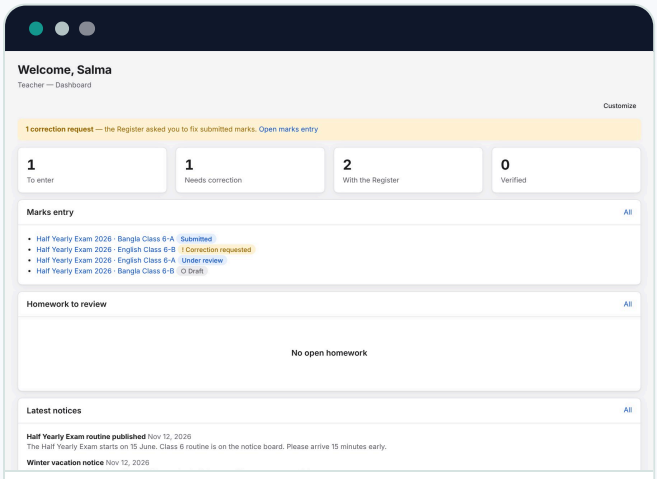
## Teacher

- Sees **only their own** classes and subjects
- Attendance, marks entry, class tests, homework
- Submits marks to the Register; fixes corrections
- Messages colleagues, own class group

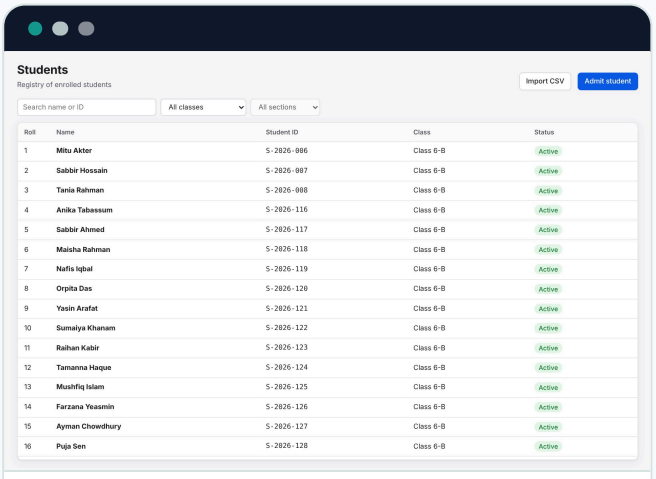
## Student

- Sees **only their own** record
- Published results and report cards
- Attendance, class tests, homework, notices
- Messages to their own teachers (school-controlled)

**Headmaster and Register are deliberately different.** The Register prepares and verifies; the Headmaster approves. The person who verified marks cannot also approve them, and approving or publishing needs the password to be typed again.




Teacher dashboard: a correction request from the Register is highlighted.




Register: student registry with search and class filters.

A DAY IN THE LIFE

# One school day, seen by each person

An illustrative day, using the demonstration school. It shows how the pieces fit together without anyone retyping anything.

**8:05 · TEACHER**

### Takes attendance in seconds

Roll call on a tablet or computer; absences are counted straight away.

**10:30 · TEACHER**

### Gives homework

Posted to the class; students see it in the app.

**12:00 · TEACHER**

### Submits exam marks

Range-checked, totalled; locked once submitted.

**13:15 · REGISTER**

### Checks the submission

Spots a doubtful total and requests a correction with a reason.

**14:00 · TEACHER**

### Fixes and resubmits

Saved as a new revision; the old one is kept.

**15:30 · HEADMASTER**

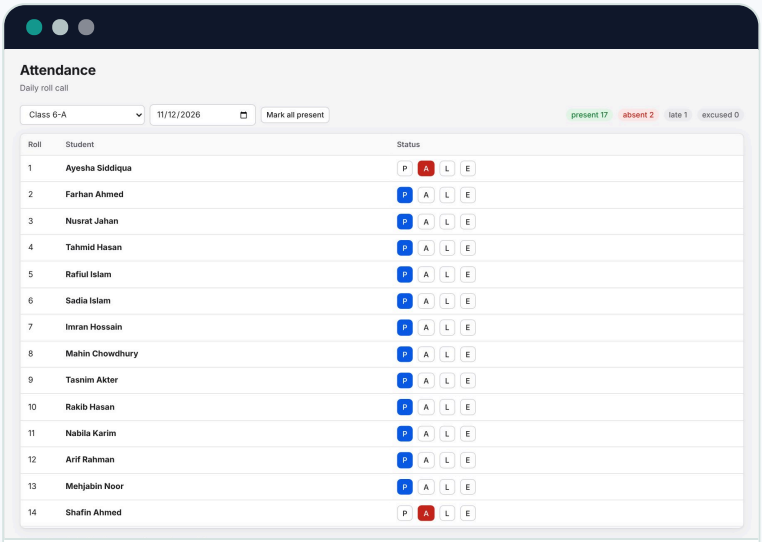
### Approves and publishes

Password typed again; the decision is recorded.

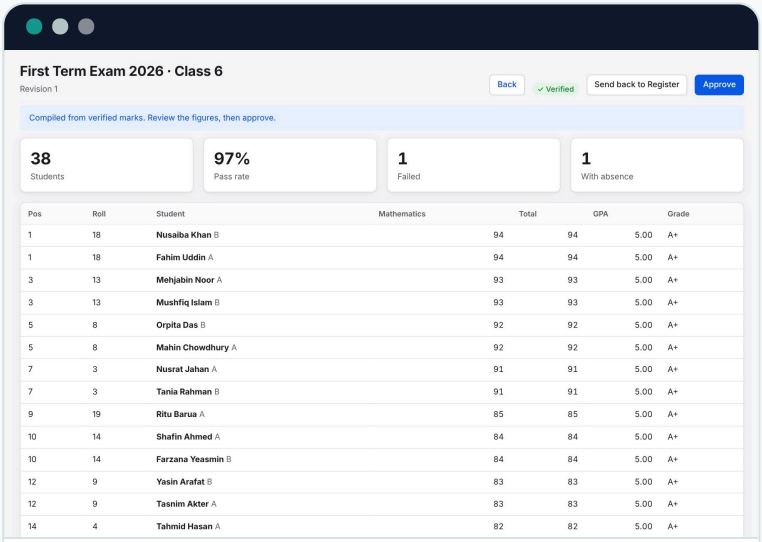
**15:31 · STUDENT**

### Sees the result

A notification arrives; only their own result opens.




Teacher: roll call with present, absent and late.




Headmaster: approval screen with the full result before publishing.

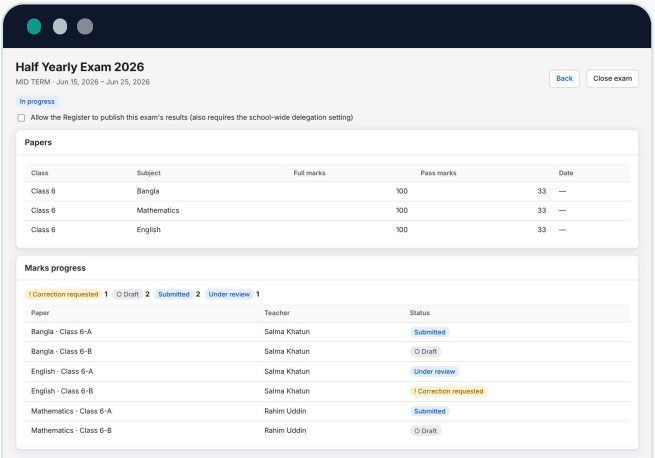
**Result:** the work that used to take days of sheets and walking between rooms becomes a short chain of clear steps — and each step is visible to the next person.

THE MOST IMPORTANT PROCESS

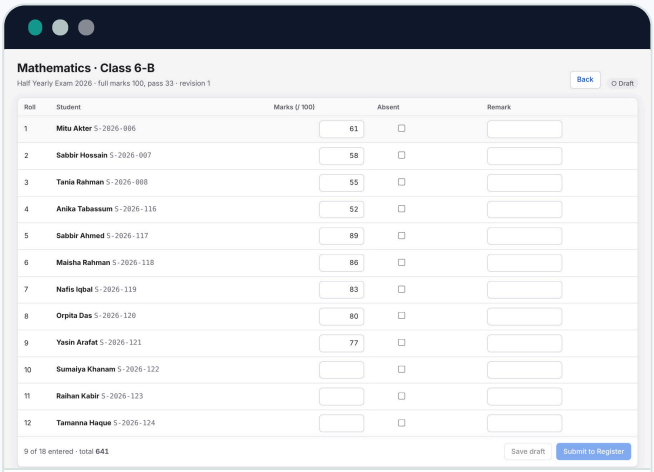
# From marks to published result

Results are the most sensitive thing a school publishes. ShikkhaOS makes the path explicit, so every step has an owner and a record.

1. **Register creates the exam**
   Adds papers for each class and subject.
2. **Headmaster authorizes**
   Marks entry opens for the assigned teachers.
3. **Teacher enters marks**
   Invalid marks are rejected on the spot.
4. **Teacher submits**
   The sheet is locked — no silent edits.
5. **Register reviews**
   Can request a correction with a reason.
6. **Register verifies**
   Then compiles the class result.
7. **Headmaster approves**
   Password confirmed again. Can send back.
8. **Result is published**
   Only after approval. Cannot be overwritten.
9. **Students are notified**
   Each sees only their own result.




Exam progress board: every paper and section with its current status.




Teacher marks entry: keyboard-friendly grid, running total, save draft / submit.

**Corrections are visible.** The Register flags rows and writes a reason; the teacher sees it and resubmits as a new revision. Old versions are kept.

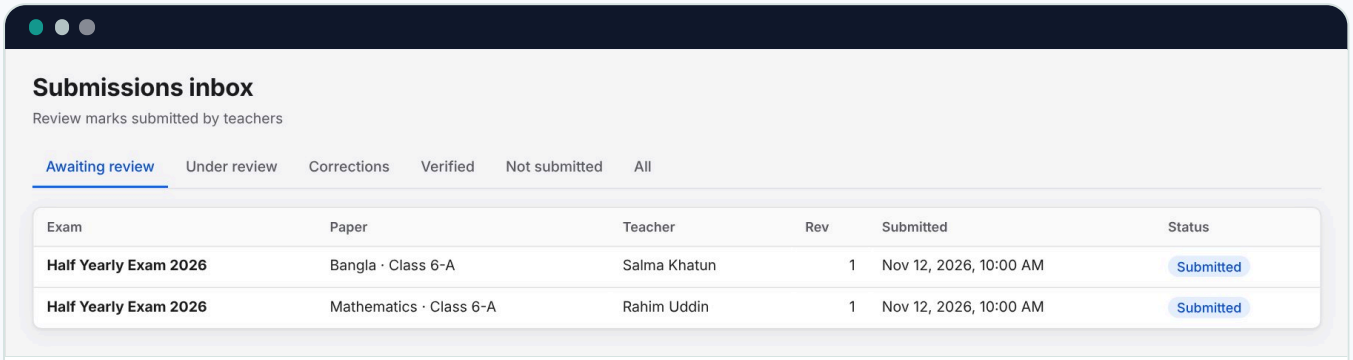
**Published means final.** A published result cannot be edited. A mistake found later needs a request approved by the Headmaster and creates a new revision — the old one stays on record.

**Class tests are separate.** Teachers can share class-test marks with their class, but these never change the official result.

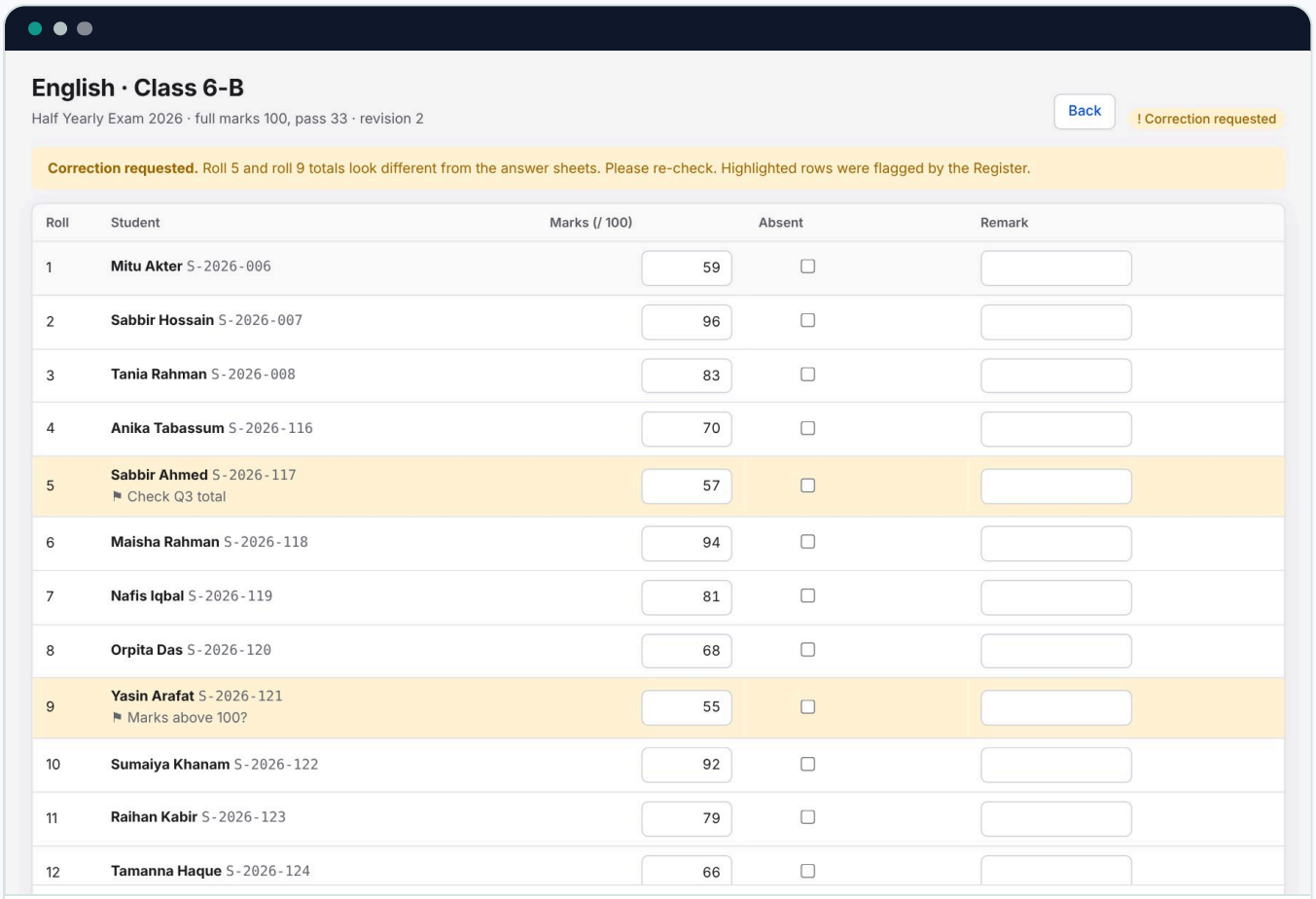
IN USE — REVIEW

# The Register checks every submission

Submitted marks arrive in the Register's inbox. Nothing is lost between the teacher and the Register, and nothing can be changed quietly.




Register's inbox: submissions waiting for review, filtered by status (awaiting review, under review, corrections, verified).




Teacher's view after a correction request: the Register's reason and the flagged rows are shown; the teacher fixes and resubmits as a new revision.

**Clear statuses**
Draft, Submitted, Under review, Correction requested, Verified — always visible.

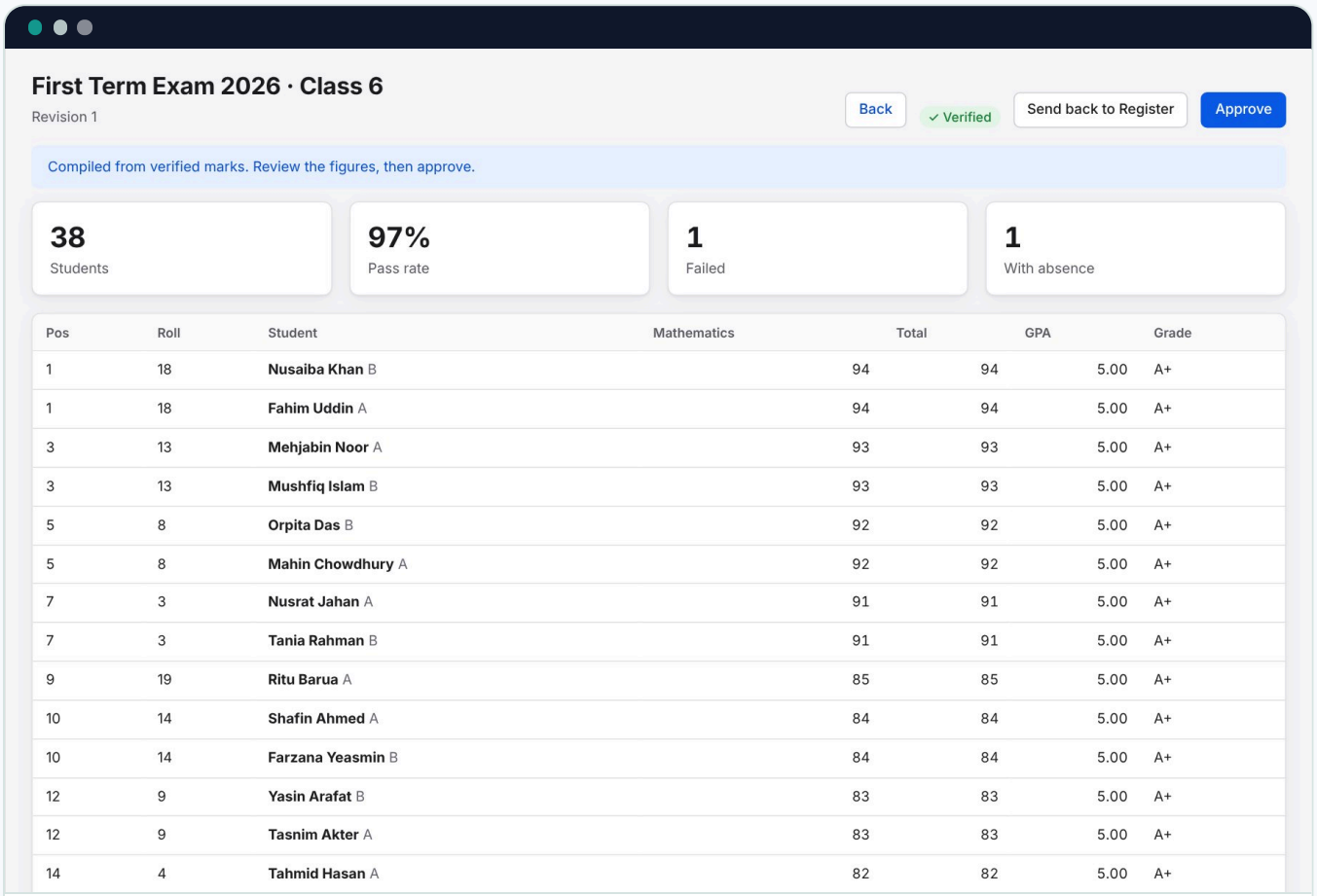
**Reasons are required**
A correction request must say why; the teacher sees it next to the marks.

**History is kept**
Each submission keeps every revision and who did what.

IN USE — APPROVAL

# The Headmaster decides

Once every paper of a class is verified, the Register compiles the class result. The Headmaster sees the numbers and approves — or sends it back.




| Pos | Roll | Student | Mathematics | Total | GPA | Grade |
|---|---|---|---|---|---|---|
| 1 | 18 | **Nusaiba Khan** B | 94 | 94 | 5.00 | A+ |
| 1 | 18 | **Fahim Uddin** A | 94 | 94 | 5.00 | A+ |
| 3 | 13 | **Mehjabin Noor** A | 93 | 93 | 5.00 | A+ |
| 3 | 13 | **Mushfiq Islam** B | 93 | 93 | 5.00 | A+ |
| 5 | 8 | **Orpita Das** B | 92 | 92 | 5.00 | A+ |
| 5 | 8 | **Mahin Chowdhury** A | 92 | 92 | 5.00 | A+ |
| 7 | 3 | **Nusrat Jahan** A | 91 | 91 | 5.00 | A+ |
| 7 | 3 | **Tania Rahman** B | 91 | 91 | 5.00 | A+ |
| 9 | 19 | **Ritu Barua** A | 85 | 85 | 5.00 | A+ |
| 10 | 14 | **Shafin Ahmed** A | 84 | 84 | 5.00 | A+ |
| 10 | 14 | **Farzana Yeasmin** B | 84 | 84 | 5.00 | A+ |
| 12 | 9 | **Yasin Arafat** B | 83 | 83 | 5.00 | A+ |
| 12 | 9 | **Tasnim Akter** A | 83 | 83 | 5.00 | A+ |
| 14 | 4 | **Tahmid Hasan** A | 82 | 82 | 5.00 | A+ |

Headmaster's approval screen: pass rate, failures and absences at a glance. Actions: Approve, or Send back to Register with a reason.

**Fewer surprises**
Pass rate, failures, absences and every student's marks, grade and position are shown before approval.

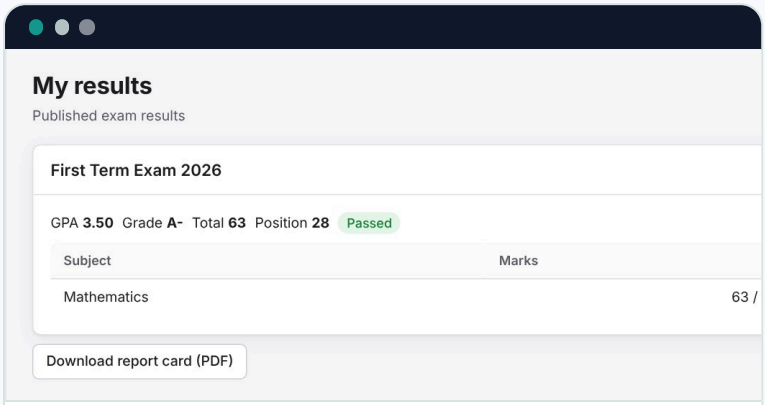
**Send back**
If something looks wrong, the Headmaster returns the result with a reason. It is re-checked and re-compiled as a new revision.

**Two-person control**
The person who entered marks cannot verify them; the verifier cannot approve them. Approval needs the password again.

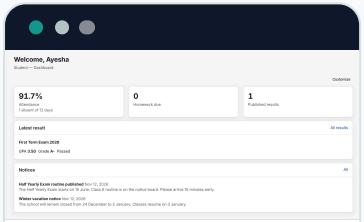
IN USE — STUDENTS

# Students see their own results — and only theirs

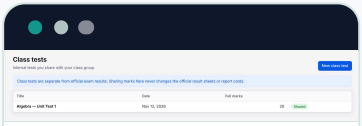
When the Headmaster publishes, each student is notified and can open their result and download a report card.



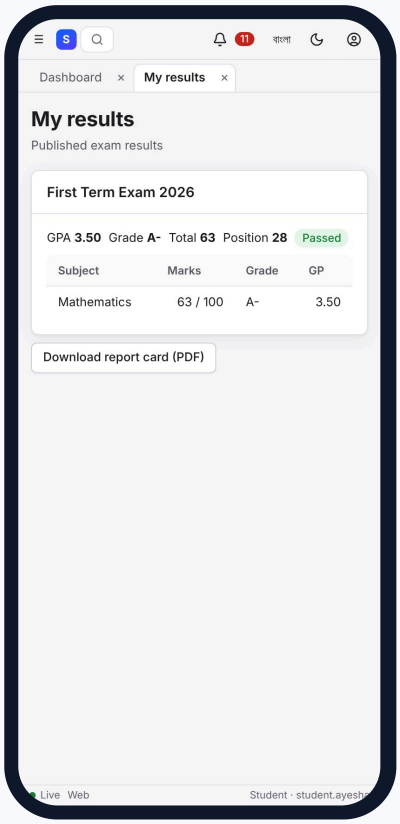

Student's result page (web): subject-wise marks, grade and GPA.




Student dashboard: attendance, homework, latest result, notices.




Class tests: shared with the class, clearly separate from official results.




Result on a phone

**Same rules everywhere.** A student can open only their own result, attendance and homework — on the phone, the web or any other device.

**Note.** The phone pictures show the student screens on a phone-sized display. The dedicated Android app has its own native screens with the same features; it is built and tested, and its first installable build is produced by the automated release process (see “Where we are today”).

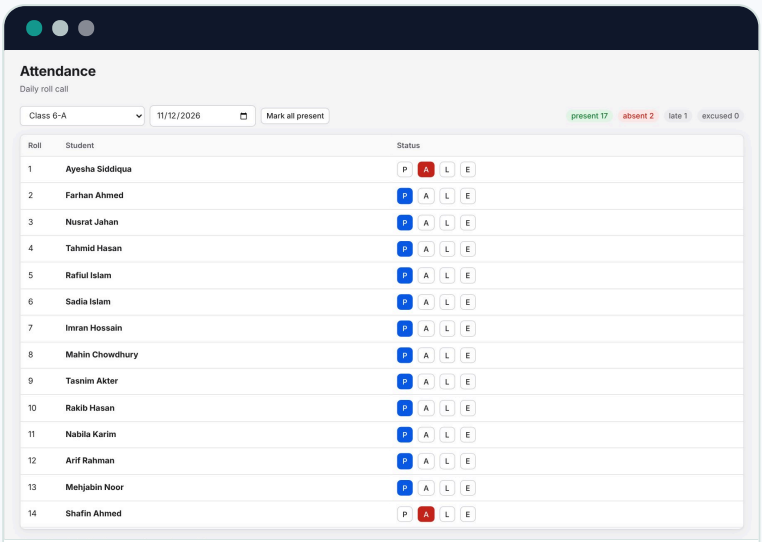
IN USE — DAILY SCHOOL LIFE

# Attendance, class tests and homework

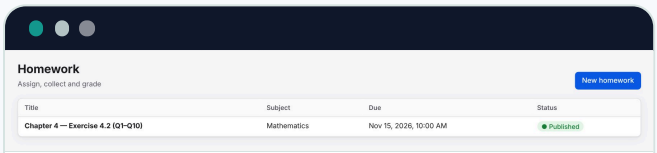
## Attendance in seconds

The class teacher marks Present / Absent / Late / Excused for the whole class on one screen. Students marked absent are notified. Past records can be corrected only with a written reason, and the correction is logged.

- ✓ “Mark all present” then change exceptions
- ✓ Attendance % per student and per section
- ✓ Summary report for the Headmaster




Teacher: roll call for Class 6-A.




Teacher: homework with due date; students submit text or photos/files; teacher grades with feedback.

## Homework flow

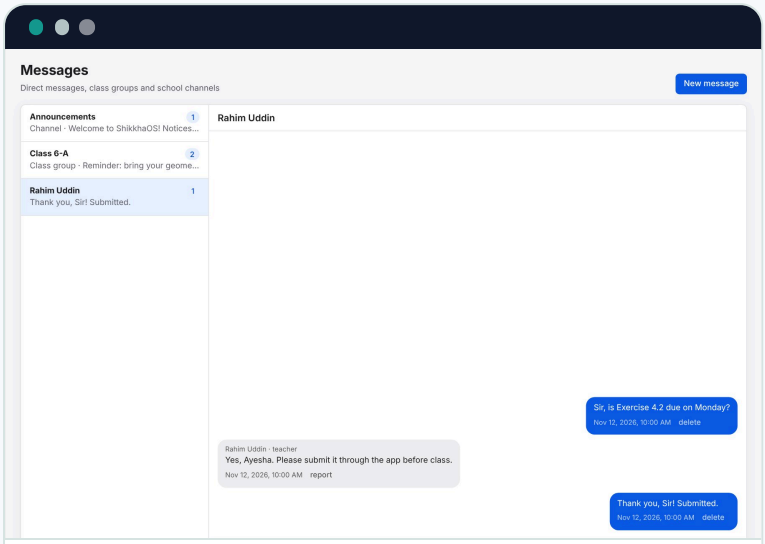
1. Teacher creates and publishes homework for a class.
2. Students are notified, and reminded 24 hours before the deadline.
3. Students submit answers (text, photo or file).
4. Teacher grades and gives feedback; the student is notified.

Late submissions are flagged automatically.

COMMUNICATION AND STUDENT SAFETY

# Messages the school can stand behind

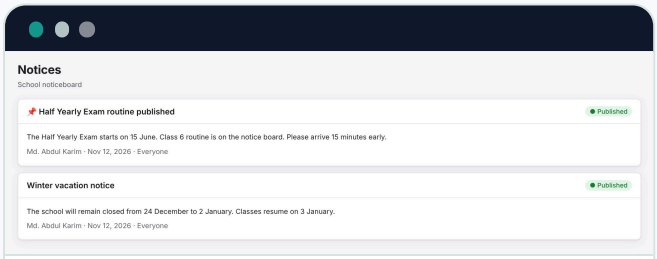
Communication is built into ShikkhaOS so that it follows school rules and protects students.



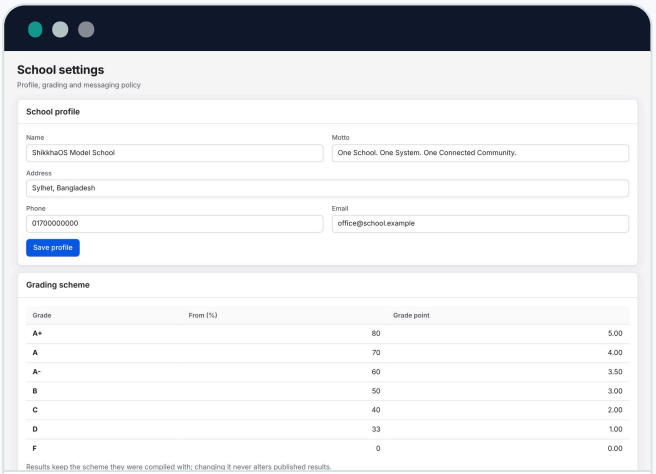

A student asks her teacher a question; the teacher replies. Messages arrive live.

## Safety rules built in

- **No private chat between students.** It is switched off and cannot be enabled by mistake.
- Students can message only their **own teachers** and the school office, and the school can switch either off.
- Class groups follow enrollment automatically; students read, and post only if the teacher allows.
- Any message can be **reported**; the Headmaster can hide it or suspend a user's messaging.
- When a moderator reads a student conversation, **that read is recorded**.



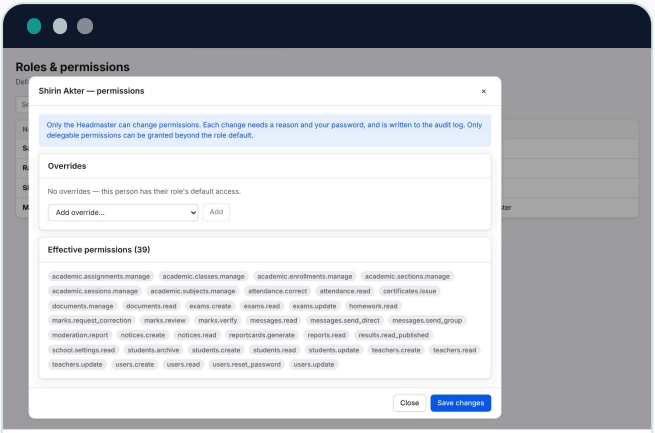

Notices: targeted to everyone, a role, a class or a section; pinned notices stay on top.



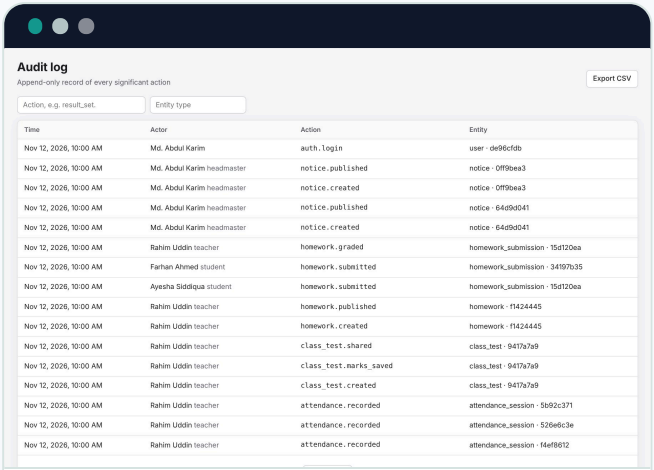

School settings: the Headmaster controls messaging and publishing policy.

CONTROL, TRUST AND RECORDS

# Who can do what — and a record of everything



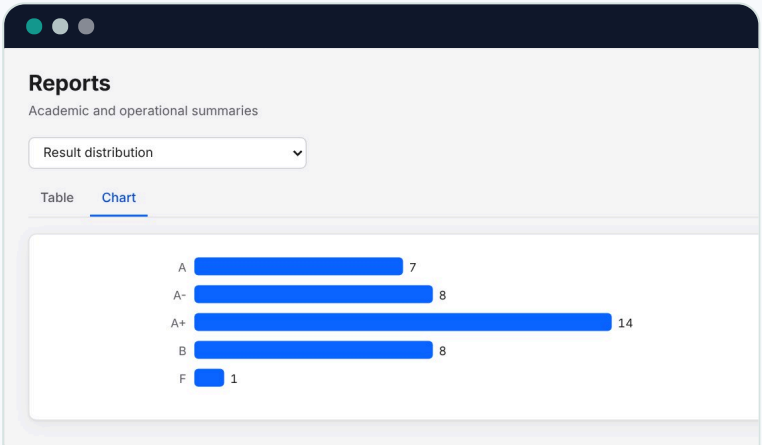

Headmaster: per-person permissions. Every change needs a reason and the password.




Audit log: a permanent, read-only history of important actions.

## Protection at a glance

- Passwords stored securely; account locks after repeated wrong attempts.
- Staff and students see only what their role allows.
- Approving or publishing needs the password typed again.
- Published results and the audit log cannot be altered.
- Uploaded files are checked before they are accepted.
- Private information is never put in notification text.




Reports with charts (and a table alternative): grade distribution, attendance, enrollment, exam analysis.

TRUST AND SECURITY

# How student data and results are protected

Protection is enforced by the school server — the screens only reflect it. This table lists what could go wrong and what stops it.

| What could go wrong | What prevents it |
|---|---|
| Someone guesses or steals a password | Passwords are stored in a one-way secure form; repeated wrong attempts lock the account; sessions expire and can be ended per device. |
| A staff member views or edits what they should not | Every request is checked against the person's role and assignment on the server. A teacher cannot open another teacher's class. |
| A mark is quietly changed after submission | Submitted marks are locked. Corrections require a reason and create a new revision; the audit log records who and when. |
| One person verifies and approves their own work | Separation of duties is enforced: the verifier cannot approve. Approving and publishing need the password again. |
| A published result is altered later | Published results are protected at the database level; changes appear only as a new, visible revision. |
| Students contact each other privately or are contacted by strangers | Student-to-student chat does not exist. Messages are limited to their own teachers and the office; any message can be reported and hidden. |
| An unsafe file is uploaded | Files are checked for type and size before being accepted. |
| Private data leaks through notifications | Notification text never contains marks or private details. |

### Checked, not just promised

The server's security rules are covered by automated tests, a dependency vulnerability audit passes, and the screens were checked against accessibility standards (WCAG 2.1 A/AA).

### Still the school’s decision

Where the data is hosted, who holds admin passwords, and how often backups are checked are policy choices for the committee — see the decisions page.

WHERE WE ARE TODAY

# What is built, tested — and what remains

## Built and tested

- Complete school server and all four user roles.
- Exam → marks → review → approval → publish workflow, including corrections, send-back and revisions.
- Attendance, class tests, homework, notices, messaging with safety rules, reports, certificates and report cards (PDF).
- Web app, Windows desktop app and Android student app (source complete).
- Bulk student import from a spreadsheet (CSV).
- English and Bengali (core screens on web/desktop).

## Quality checks run so far

- 48 automated tests on the server (including full result workflow and security checks).
- 30 tests on the apps, plus 8 end-to-end tests that click through the real screens.
- Accessibility check (WCAG 2.1 A/AA) on about 20 screens: no violations found.
- A full exam cycle was run end to end through the real screens.

## Still to do before go-live

| Item | Why it is open |
|---|---|
| Windows installer and Android app files | Produced by the automated release process on GitHub; their first run and signing keys are still to be set up and tested. |
| Phone push notifications | In-app and live notifications work; delivery to a closed phone needs a Firebase setup. |
| Hosting, backups and monitoring | Depends on the committee's hosting choice (next pages). |
| Android app in Bengali | Web/desktop are bilingual; the Android app is English-only for now. |
| Bengali text on printed PDFs | Report-card PDFs currently support English text only; Bengali names need a font upgrade. |
| Real-school pilot | The system has been tested with demonstration data, not yet with our real classes. |

ROLLOUT

# A safe, step-by-step start

We recommend starting small, learning, and expanding. Durations are indicative estimates to be confirmed with the committee.

PHASE 1 · ABOUT 2 WEEKS

### Prepare

- Choose hosting; set up the school server and backups
- Create Headmaster, Register and teacher accounts
- Import students from a spreadsheet

PHASE 2 · ABOUT 4 WEEKS

### Pilot: one class

- Attendance, homework, notices, messages
- Short training for the pilot teachers
- Weekly feedback meeting

PHASE 3 · ONE EXAM CYCLE

### First exam on the system

- Marks entry, review and approval for the pilot class
- Run alongside the usual paper process once
- Compare results; fix issues

PHASE 4 · AFTER REVIEW

### Whole school

- All classes and teachers
- Student Android app rollout
- Ongoing support and updates

## What running it needs

- A small server (school office computer or rented cloud server), always on, with internet.
- A web address (domain) and a security certificate for the site.
- Daily automatic backups, stored away from the server.
- One person per week for the first months for support questions.

An itemized cost estimate will be prepared once the hosting option is chosen.

## Risks and how we handle them

| | |
|---|---|
| **Staff unfamiliar** | Short hands-on training, pilot first. |
| **Internet outage** | Paper fallback during pilot; apps recover automatically. |
| **Data loss** | Automatic backups and tested restore. |
| **Wrong access** | Roles reviewed by the Headmaster; audit log. |

FOR THE COMMITTEE

# Decisions we request

Six decisions allow the pilot to begin.

- [ ] **Approve the pilot** in one class (suggested: Class 6, two sections) for one term, as described in Phases 1–3.
- [ ] **Choose where the system runs:** (a) rented cloud server, or (b) a school-owned server. We will prepare costs for the chosen option.
- [ ] **Decide who may publish results.** Default recommendation: **only the Headmaster**. The Register may be allowed per exam if the committee wishes.
- [ ] **Nominate a system owner** (suggested: the Register) and set aside training time for pilot teachers.
- [ ] **Confirm messaging policy:** students may message their own teachers and the school office (recommended), or teachers only.
- [ ] **Confirm data handling:** student records stay on the school's own system; who may export data and how often backups are checked.

## What the school provides

- List of students (name, ID, class, section, roll, guardian phone).
- List of teachers and their class/subject assignments.
- Grading rules (current marks-to-grade scale).
- A decision-maker for questions during the pilot.

## What the committee receives

- A working system for the pilot, with training.
- Weekly progress notes during the pilot.
- A pilot review: what worked, what to change, whether to proceed.
- Full source and documentation for the school's records.

**Committee decision**
Date: ___ / ___ / _____

Approved ☐ Approved with changes ☐ Not approved ☐
Chairperson signature: _____

APPENDIX

# Everything the system covers

| Area | What it includes |
|---|---|
| Identity and access | Sign-in, forced password change, device list, per-person permissions, password re-confirmation for sensitive actions. |
| School setup | School profile, grading scale, academic sessions, classes, sections, subjects, teacher assignments, promotions. |
| People | Student registry (with CSV import and archive), teacher records, enrollments, class groups. |
| Examinations | Exams and papers, authorization, marks entry, submission, review, correction, verification, compilation, approval, send-back, publishing, revisions. |
| Daily work | Attendance, class tests (separate from official results), homework and grading. |
| Communication | Direct messages, class groups, official channels, notices with targeting, notifications, moderation. |
| Documents | Report cards and certificates (PDF), document library, file attachments with safety checks. |
| Reports and oversight | Enrollment, attendance, exam analysis, teacher submission status, grade distribution; CSV export; audit log; global search. |

## Technology, in plain words

The school server is written in Rust, a language known for speed and safety, and stores data in PostgreSQL, a widely trusted database. The apps use common, well-supported tools. Updates are delivered as versioned releases with checksums so the school can verify them.

## Document information

Prepared from the working system and its automated test results. Screens show demonstration data. Quality figures are those recorded at the time of writing (November 2026).

# Let’s begin with one class.

A small pilot lets teachers, the Register and the Headmaster judge ShikkhaOS on real work — with paper as a safety net — before the whole school commits.

**Step 1**
Committee approves the pilot

**Step 2**
We prepare the server, accounts and training

**Step 3**
One class, one term, weekly review

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